

SmartCare Complete Review Report

After reviewing **31 backend files** and **12 frontend files/components**, I can summarize the situation like this:

The database design is actually strong.

The backend architecture is good.

The frontend workflow does not fully reflect the SmartCare concept.

The biggest problem is that the system was designed around:

Assessment

↓

Priority Determination

↓

Appointment

↓

Queue

but the implementation currently behaves more like:

Register

↓

Book Appointment

↓

Queue

Priority 1: Critical Security Fixes

These should be fixed before anything else.

1. Registration Allows Users To Become Admins

Current

Register page:

Patient

Doctor

Nurse

Receptionist

Admin

Backend:

```
User.create({  
  email,  
  password_hash,  
  role  
})
```

Why It's Bad

Anyone can register as:

ADMIN

DOCTOR

NURSE

which completely breaks the security model.

Fix

Remove role selection from registration.

Registration should always:

Create PATIENT only

Staff and Admin accounts should be created by Admin users.

2. Password Hash Exposure

Current

User Controller returns:

```
res.json(user)
```

which includes:

```
password_hash
```

Why It's Bad

Even hashed passwords should never be sent to the frontend.

Fix

Exclude:

```
password_hash
```

from all responses.

3. User Information Exposure

Current

Any logged-in user can access:

```
GET /users/:id
```

Why It's Bad

Patients can potentially retrieve other user records.

Fix

Only:

```
ADMIN
```

or the account owner should view user records.

4. Staff Information Exposure

Current

```
GET /staff/:id
```

has no ownership protection.

Fix

Restrict access to:

ADMIN

Healthcare Staff

or owners only.

5. Queue Information Exposure**Current**

GET /queue/:id

lacks ownership validation.

Fix

Patients should only see:

Their own queue entry

6. Patient List Exposure**Current**

Patient.findAll()

can be requested by any authenticated user.

Fix

Restrict to:

ADMIN

DOCTOR

NURSE

RECEPTIONIST

``

Priority 2: SmartCare Business Logic Missing

These are the biggest project-design issues.

7. Assessment Is Not Driving The System

Current

Patient can go straight to:

Book Appointment

Why It's Bad

The SmartCare innovation disappears.

Fix

Force workflow:

Register

↓

Complete Assessment

↓

Priority Generated

↓

Appointment Booking

8. Patients Choose Their Own Priority

Current

Appointment Page:

Normal

High

dropdown.

Why It's Bad

Patients should never decide urgency.

Fix

Remove priority selector completely.

Priority should come from:

Assessment Results

9. Assessment Priority Is Client-Controlled

Current

Frontend sends:

priority_level

Backend stores it.

Fix

Backend must calculate:

HIGH

MEDIUM

STANDARD

from symptoms.

10. Recommended Service Is Client-Controlled

Current

Frontend can send:

Emergency

Pharmacy

Fix

Backend should determine service.

Example:

Chest Pain

↓

Emergency

11. Appointment Doesn't Require Assessment

Current

Users can create appointments without:

assessment_id

..

Fix

Require valid assessment before appointment creation.

12. Department Selection Ignores Assessment Results

Current

Patient chooses department.

Fix

Assessment should suggest or automatically assign:

Emergency

Pharmacy

General Practice

based on symptoms.

Priority 3: Queue System Needs Rework

The queue is currently CRUD rather than Smart.

13. Queue Priority Is Client-Controlled

Current

Frontend sends:

priority_score

Fix

Backend calculates score.

14. Queue Number Is Client-Controlled

Current

Frontend provides:

Queue #22

Fix

Backend generates queue numbers.

15. No Queue Prioritization Logic

Current

Queue stores data.

Missing

Highest Priority First

logic.

Fix

Sort queue:

Priority Score DESC

and serve patients accordingly.

Priority 4: Registration & Profile Workflow

16. Registration Doesn't Create Complete Patient Profile

Current

Creates:

User

only.

Missing

Patient Profile

..

Fix

Immediately collect:

First Name

Last Name

DOB

Phone

or redirect to profile completion.

17. Registration Form Is Too Basic

Current

Email

Password

only.

Fix

Create a proper patient onboarding process.

Priority 5: Frontend Redesign

Landing Page

18. Doesn't Sell SmartCare

Users only see:

Book appointments

Add

Symptom Assessment

Priority Scheduling

Smart Queue Management

``

19. No Workflow Explanation

Add:

1. Register

2. Assessment

3. Priority Level

4.

20. No Trust Indicators

Add:

Secure Records

Fast Triage

Queue Monitoring

Login Page

21. No SmartCare Identity

Looks like generic login.

Add healthcare branding.

22. Patient Redirect Is Wrong

Current:

Login

↓

Appointments

Should be:

Login

↓

Dashboard

23. No Forgot Password

Add later.

Appointment Page

24. Most Important Page To Redesign

Remove:

Priority selection

Add:

Assessment result summary

Recommended service

Priority level

before booking.

Staff Dashboard

25. Displays UUIDs

Current:

Department ID

Appointment ID

Replace With

Patient Name

Department Name

26. Not Prioritization Focused

Make high-priority patients visually dominant.

27. Missing Dashboard Cards

Show:

Patients Waiting

High Priority Patients

Appointments Today

28. Missing Filters

Add:

High Priority

Department

Status

Admin Dashboard

29. Isn't Actually A Dashboard

Current:

Patient List

``

Should Display

Total Patients

Appointments

Queue Size

Staff Count

Departments

Navbar

30. Wrong Patient Navigation

Current:

Book Appointment

Should Be

Assessment

Appointments

Queue

Notifications

31. Missing Queue Link

Patients should track queue position.

32. Missing Notifications Link

Notifications currently feel invisible.

Priority 6: Backend Architecture Improvements

33. Missing Sequelize Relationships

Add:

Assessment ↔ Appointment

Department ↔ Appointment

Staff ↔ Appointment

34. Appointment Missing assigned_staff_id

Required for:

Doctor Assignments

Staff Dashboards

35. Notification Model Doesn't Fully Match Schema

Review:

appointment_id

sent_at

support.

36. AuthContext Improvements

Add:

logout()

inside context.

37. Better Error Pages

Add:

Retry Button

Back To Dashboard

instead of only showing errors.

Final Verdict

If I were rebuilding SmartCare in order, I'd do:

Phase 1 (Must Fix)

1. Registration role vulnerability
2. User/staff/patient privacy issues
3. Password exposure
4. Assessment required before appointment

Phase 2 (Core SmartCare)

5. Backend priority calculation
6. Backend recommended service calculation
7. Queue prioritization engine
8. Queue number generation

Phase 3 (Frontend Redesign)

9. Appointment page redesign
10. Register page redesign
11. Staff dashboard redesign
12. Admin dashboard redesign

13. Navbar redesign

Phase 4 (Enhancements)

14. Notifications center

15. Queue tracking page

16. Profile completion workflow

17. Analytics & reporting

Overall Project Score (Current)

Area	Score
Database Design	8.5/10
Backend Structure	8/10
Backend Security	5/10
Smart Prioritization Logic	4/10
Frontend UI	6/10
Frontend Workflow	4/10
Final Year Project Potential	8.5/10